





Age of marriage of women in Rajasthan

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Introduction

Does caste affect the age of marriage of women in Rajasthan? Or is education the real determinant, while caste is just a proxy?

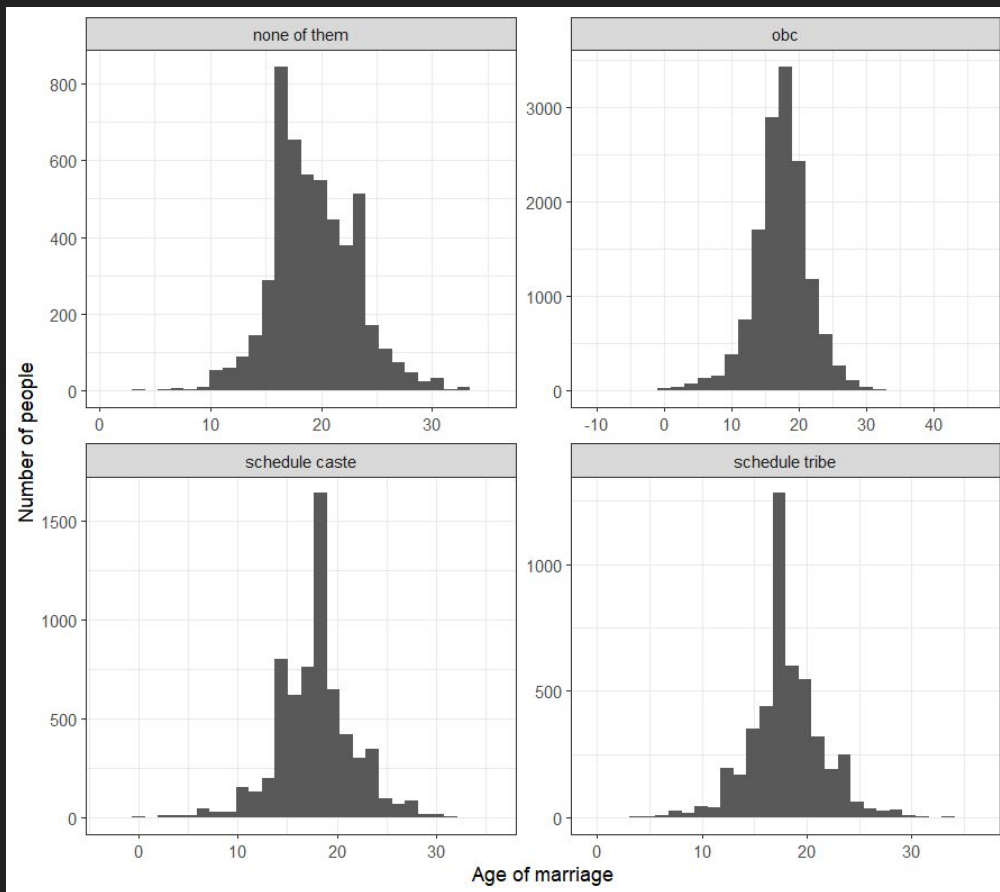
Data

- Data Considered : National Family Health Survey -5 (2019-21)
 - 5,972 columns
 - Approximately 1.8 million people
- Data Cleaning
 - Using STATA
 - 5,072 variables and 42,990 people
 - 5,072 variables and 30,566 people

Summary statistics of marriage age by caste

Caste	Mean	Standard deviation	Sample size
None of them	19.4	3.8	5095
OBC	17.9	4.1	14260
SC	17.9	3.9	6465
ST	18.2	3.7	4678

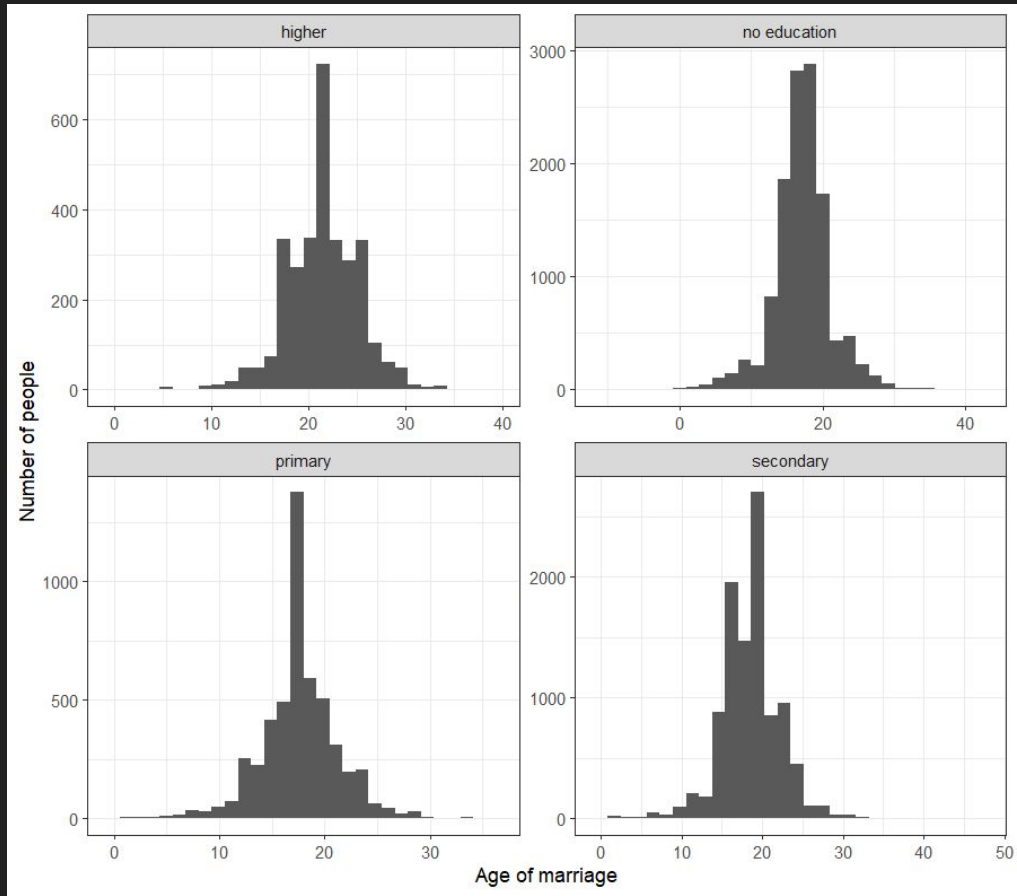
Table -1 : Mean, Standard Deviation and Sample size of the SC, ST, OBC and 'None of them' groups



Summary statistics of marriage age by education

Education	Mean	Standard deviation	Sample size
Higher	21.5	3.8	3082
Secondary	18.6	3.6	10174
Primary	17.7	3.8	4988
No education	17.3	4.02	12322

Table -1 : Mean, Standard Deviation and Sample sizes of the Higher, Secondary, Primary and No Education Groups



Hypothesis Testing

$$H_0 : \mu_1 = \mu_2$$

$H_a : \mu_1 \neq \mu_2$, where, μ_1 and μ_2 are the mean of sample groups 1 and 2

- Two sample Z-test as sample size of the groups is greater than 30
- 5 % significance level
- Z test statistics:

$$\frac{\mu_1 - \mu_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Hypothesis Testing (Caste)

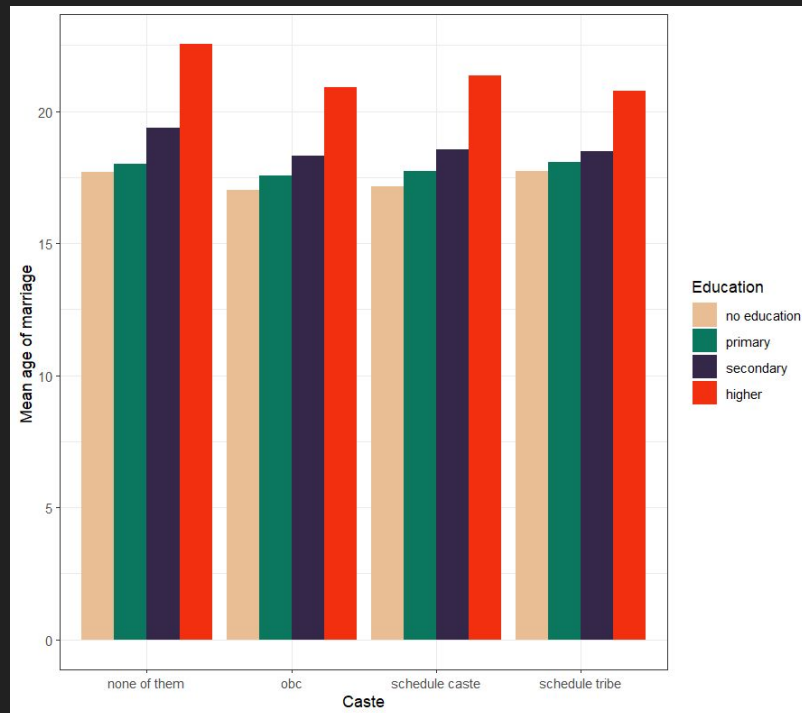
Is there a significant difference in age of marriage for women across different caste groups?

Categories	Z statistics	Hypothesis Testing
None of them - SC	20.8	Reject
None of them - ST	15.8	Reject
None of them - OBC	24.19	Reject
OBC - SC	0	Fail to Reject
OBC - ST	-4.7	Reject

Hypothesis Testing (EDUCATION)

Is there a significant difference in age of marriage for women across different level of education?

Categories	Z statistic	Hypothesis Testing
Higher - No Education	54.2	Reject
Higher - Primary	43.7	Reject
Higher - Secondary	37.6	Reject
No Education- Primary	-6.2	Reject
No Education - Secondary	-25.6	Reject
Primary - Secondary	-13.9	Reject



	higher	secondary	primary	no education
none of them	22.56152	19.39197	17.99735	17.68883
obc	20.89862	18.30754	17.56351	17.00597
schedule caste	21.34343	18.54413	17.75202	17.17195
schedule tribe	20.78292	18.49345	18.09393	17.72578

Limitations

- Data quality issues:
 - Some entries had invalid values (e.g., negative ages).
 - Age at marriage was not directly provided. we had to calculate it using birth year and year of marriage, which may introduce errors.
- Causality cannot be established, only correlation observed.
- Urban–rural differences were not explored in this analysis.

Conclusion

- Caste shows a statistically significant association with women's age at marriage in Rajasthan.
- However, education level has a much stronger and more consistent effect.
- Women with higher education marry significantly later than those with no education.
- Caste may influence age indirectly through access to education and socio-economic conditions.
- Promoting girls' education may be more effective than caste-targeted efforts in delaying marriage age.

Thank You!